

Letter to the Editor

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Letter by Frösen et al Regarding Article “Potential Influences of Gut Microbiota on the Formation of Intracranial Aneurysm”

To the Editor:

It is with great excitement that we read the recent article entitled Potential Influences of Gut Microbiota on the Formation of Intracranial Aneurysm by Shikata et al.¹ In this study, Shikata et al¹ demonstrated in a mouse model of intracranial aneurysm (IA) formation induced by intrathecal elastase injection coupled with hypertension, how depletion of gut microbiota affects the formation of IAs.¹ This effect was related to reduced inflammation at the site of IA formation in the cerebral artery wall.¹ Inflammation of the artery and aneurysm wall has been shown in several studies to be a driver of IA initiation, progression, and eventual rupture.² The study by Shikata et al¹ introduces a novel factor modulating or perhaps even causing this inflammation.

While experimental models such as the one used by Shikata et al¹ are invaluable to build our understanding of the mechanisms mediating IA formation, growth, and rupture, one has to always bear in mind that none of the currently available models reconstitutes the multifactorial pathobiology of IA formation, growth, and rupture in humans.^{2,3} This has important implications for the extrapolation of the experimental results into humans and ultimately patient care. Inspired by the results of Shikata et al,¹ who used a per oral antibiotic regimen consisting of vancomycin, metronidazole, ampicillin, and neomycin to deplete gut microbiota, we looked in our previously described population-based registry-derived cohort of patients with de novo IA formation⁴ whether use of antibiotics was associated with risk of de novo IA formation. This was done by combining data obtained from a national registry for prescribed medications with the data from our population-based IA registry. To our surprise, patients with documented de novo IA formation had used significantly more antibiotics than those without de novo IA formation during follow-up (Table).

The apparently contradictory results obtained from our patient cohort demonstrate the challenges of extrapolating observations from highly controlled mouse models to actual patients with varying genomics, exposome, therapeutics, treatment compliance, and among many other things—microbiota. The observation reported by Shikata et al¹ is nevertheless of great value. Combined with our data, it suggests that it might be the variations in microbiota or microbial exposure that determine the inflammatory response to factors triggering IA formation in the cerebral artery. It is not unequivocally clear from the article by Shikata et al¹ whether they observed signs of direct bacterial transmigration to the site of aneurysm formation. Therefore, we presume that this was not the case since signs of bacterial infection or presence of bacterial components in the cerebral artery wall were not reported, although Shikata et al¹ did study

the artery wall itself. If this interpretation is correct, the study by Shikata et al¹ demonstrates how exposure to microbiota can affect the sterile inflammation occurring in distant vasculature through modulation of the immune system. Moreover, it demonstrates how modulation of the microbiota can affect the sterile inflammation occurring during IA formation in the cerebral vasculature. This, in turn, can explain our clinical observations, because the multiple antibiotic regimens prescribed to patients eventually developing de novo IAs have without a doubt had an effect also to the composition of the body's microbiota, at least temporarily. Moreover, the antibiotics were without a doubt prescribed for a reason, which, in turn, translates into higher exposure to microbial pathogens among those that developed de novo IAs (Table), an interpretation in line with the observations by Shikata et al.¹

What is very clear from the study of Shikata et al¹ is that the role of microbiota in the formation of nonmycotic IAs deserves more studies. Our observations reported above (Table) imply this as well. Of note is that the gut microbiota is not the only relevant microbiota in the body, but oral flora has been implicated in the formation and progression of IAs as well.^{5–7} In a seminal study, Pyysalo et al⁵ demonstrate the presence of DNA from oral pathogens in the wall of >50% of Finnish aneurysms that they studied. After this, they showed that IA patients have a higher incidence of severe periodontitis,⁶ demonstrating increased exposure to oral pathogens in Finnish IA patients. Subsequently, Hallikainen et al⁷ very recently confirmed this and went further by demonstrating that periodontitis and its precursor gingivitis are indeed associated with increased risk for IA formation and aneurysmal subarachnoid hemorrhage at least in the Finnish population.

Contrary to the Finnish studies, a recent study in the French population did not find traces of oral or other bacteria in the IA wall.⁸ While the role of oral pathogens in IA formation and progression may not be as important in other populations⁸ with less risk of IA rupture than the Finnish,⁹ it is also possible that the mechanism of action through which periodontal infection and oral flora affects the risk of IA formation and aneurysmal subarachnoid hemorrhage is not mediated through direct transmigration of bacterial components to the cerebral artery or IA wall. Another recent report demonstrates how swollen oral bacteria can cause modulation and dysbiosis in gut microbiota, especially in patients with periodontitis.¹⁰ Considering the findings of Shikata et al,¹ these periodontitis-related changes in gut microbiota could, in turn, explain why severe periodontitis associates with risk of IA formation and aneurysmal subarachnoid hemorrhage,^{5–7} even if traces of oral pathogens are not consistently found in the IA walls.^{5,8} For the time being, however, the pathobiological mechanisms behind the association of periodontitis, oral flora, gut microbiota, and intracranial aneurysmal disease remain to be studied and discovered. For this endeavor, of great importance are the

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Table. The amount of prescribed antibiotics used during follow-up by the KUH IA patients that developed denovo aneurysms during follow-up and those that did not

Antibiotic Use	Definition	De Novo IAs (n=42)	No De Novo IAs (n=1377)	PValue
All antibiotics				
Amount of antibiotic use	Number of purchased antibiotic packages	12 (0–72)	6 (0–229)	0.001
Length of antibiotic use	Time from first purchased antibiotic to the last one (in days)	5769 (0–7606)	3039 (0–7633)	<0.001
Length/amount of antibiotic use		437 (0–2390)	263 (0–3692)	0.002
Antibiotics commonly used for oral infections*				
Amount of antibiotic use	Number of purchased antibiotic packages	5 (0–22)	2 (0–198)	<0.001
Length of antibiotic use	Time from first purchased antibiotic to the last one (in days)	3071 (0–7500)	200 (0–7633)	<0.001
Length/amount of antibiotic use		531 (0–1701)	67 (0–3406)	<0.001

Median and range is given and Mann-Whitney U test was used for statistical comparison. Please note that the length of antibiotic use is calculated as the time from first purchase to last purchase, and does not reflect continuous antibiotic use. Therefore the ratio of time from first to last purchase and the number of antibiotic packages purchased during that time is given.

*The antibiotics commonly (but not exclusively) used to treat oral infections in Finland include: Amoxicillin, Doxycycline, Penicillin, Metronidazole, Clindamycin and Siprofloxacin.

findings by Shikata et al¹ which demonstrate how a nonconcurrent microbial exposure separated by time and location from IA formation can nevertheless have an effect on the vascular inflammation related to IA formation.¹

In summary, the experimental study by Shikata et al¹ suggests that microbiota exposure can affect the risk of IA formation, growth, and eventual rupture, while the studies by Pyysalo et al,^{5,6} Hallikainen et al,⁷ and Aboukais et al⁸ strongly suggest that this finding indeed has relevance in humans and in actual daily clinical practice. Together these studies also demonstrate the need to investigate further the role of microbiota in the complex pathobiology of IAs. The model by Shikata et al¹ offers an excellent tool for this. We enthusiastically wait for further studies on the topic.

Disclosures

None.

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